# Zero inflated models

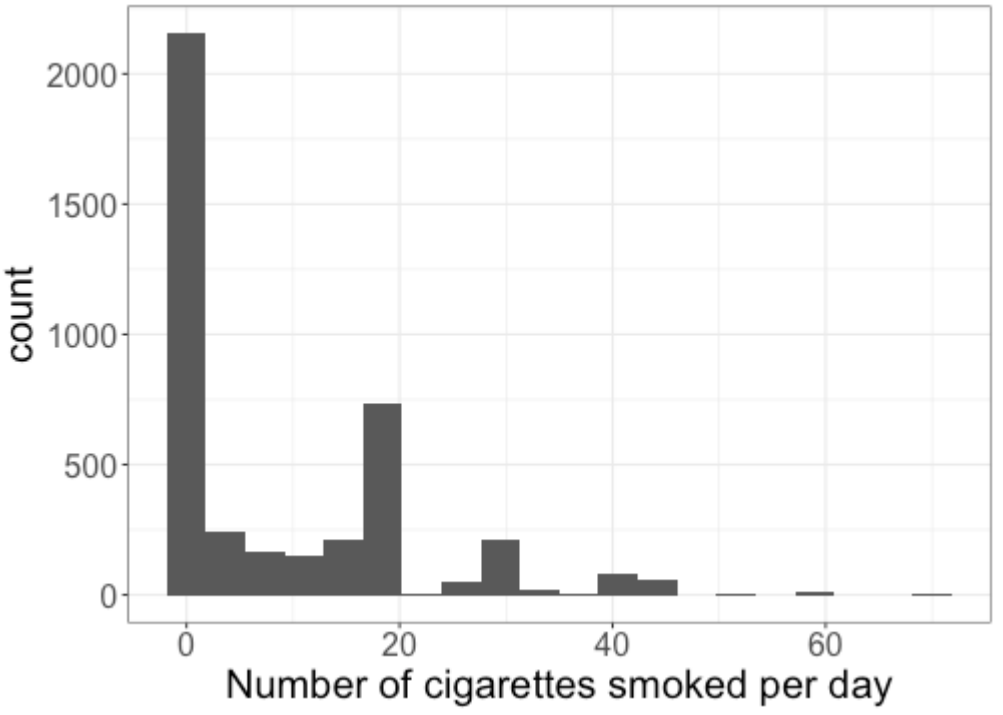
# Data: Framingham heart study

Data collected on residents of Framingham, MA over a long period of time, to study variables related to heart health. We will work with a subset of the data, containing

- `cigsPerDay`: The number of cigarettes smoked per day during the study period.
- `education`: 1 = High School, 2 = Some College, 3 = College Degree, 4 = Advanced Degree.
- `male`: 1 = Male, 0 = Female.
- `age`: The age of the individual in years.
- `diabetes`: 1 if the individual has diabetes, 0 otherwise.

> Why might we see zero inflation in the number of cigarettes smoked per day?

# EDA: number of cigarettes smoked



# Class activity

Work on the class activity (handout).

# Class activity

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1-\alpha_i) + \alpha_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\widehat{\alpha}_i}{1-\widehat{\alpha}_i}\right) = -2.51 + 0.051 Age_i$$

$$\log(\widehat{\lambda}_i) = 3.21 - 0.044 EducationSome_i - 0.082 EducationCollege_i - 0.0062 EducationAdv_i - 0.024 Diabetes_i - 0.0056 Age_i$$

> How do we interpret the coefficient -0.024 in the fitted model?

The expected # of cigarettes smoked per day for individuals who smoke is 2.4% lower for those with diabetes vs. those without diabetes, holding education, age fixed

# Class activity

$$\log\left(\frac{\widehat{\alpha}_i}{1-\widehat{\alpha}_i}\right) = -2.51 + 0.051 Age_i$$

$$\log(\widehat{\lambda}_i) = 3.21 - 0.044 EducationSome_i - 0.082 EducationCollege_i - 0.0062 EducationAdv_i - 0.024 Diabetes_i - 0.0056 Age_i$$

> What is the expected number of cigarettes smoked per day, for a 50 year old smoker with diabetes and some college education?

$\hat{\lambda}_i = \exp\{3.21 - 0.044 - 0.024 - 0.0056(50)\}$

$\approx 17.5$

# Making predictions

$$\frac{e^{-2.51 + 0.051(50)}}{1 + e^{-2.51 + 0.051(50)}} \approx 0.51$$

$$\log\left(\frac{\widehat{\alpha}_i}{1-\widehat{\alpha}_i}\right) = -2.51 + 0.051 Age_i$$

$$\log(\widehat{\lambda}_i) = 3.21 - 0.044 EducationSome_i - 0.082 EducationCollege_i - 0.0062 EducationAdv_i - 0.024 Diabetes_i - 0.0056 Age_i$$

> What if I want the expected number of cigarettes smoked per day for a 50 year old with diabetes and some college education, but *without* knowing whether they are a smoker?

If $Z_i = 1$ (nonsmoker) expected # $= 0$

If $Z_i = 0$ (smoker) expected # $= \lambda_i$

overall: expected # $= 0 \cdot P(Z_i = 1) + \lambda_i P(Z_i = 1)$

$= 0 \cdot \alpha_i + \lambda_i (1 - \alpha_i)$

$= \lambda_i (1 - \alpha_i)$

estimate: $\hat{\lambda}_i (1 - \hat{\alpha}_i) = 17.5 (1 - 0.51) \approx \boxed{18.6}$

# A new question

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1-\alpha_i) + \alpha_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\alpha_i}{1-\alpha_i}\right) = \gamma_0 + \gamma_1 Age_i$$

$$\log(\lambda_i) = \beta_0 + \beta_1 EducationSome_i + \beta_2 EducationCollege_i + \beta_3 EducationAdv_i + \beta_4 Diabetes_i + \beta_5 Age_i$$

New research question: for smokers, does the number of cigarettes smoked per day depend on age?

How would we answer this research question?

$H_0: \beta_5 = 0$ $\quad$ $H_A: \beta_5 \neq 0$

# Fitting the model in R

```
m2 <- zeroinfl(cigsPerDay ~ education +
                 diabetes + age | age,
               data = heart_data)
summary(m2)
```

```
Count model coefficients (poisson with log link):
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.2063376  0.0342290  93.673  < 2e-16 ***
education2  -0.0441182  0.0124809  -3.535 0.000408 ***
education3  -0.0820372  0.0158604  -5.172 2.31e-07 ***
education4  -0.0062453  0.0171640  -0.364 0.715964
diabetes    -0.0241459  0.0386337  -0.625 0.531974
age         -0.0056182  0.0006738  -8.338  < 2e-16 ***

Zero-inflation model coefficients (binomial with logit link):
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.506681   0.191483  -13.09   <2e-16 ***
age          0.051329   0.003821   13.43   <2e-16 ***
```

95% CI for $\beta_S$: $\hat{\beta}_S \pm 1.96\, SE(\hat{\beta}_S)$

## z-test

check whether CI contains 0

either from bootstrap resampling or R output (e.g. 0.00067)

```
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```

$H_0: \beta_S = 0$ $\quad H_A: \beta_S \neq 0$

$$Z = \frac{\hat{\beta}_S - 0}{SE(\hat{\beta}_S)} \approx -8.338 \qquad \text{p-value} \approx 0$$

# Likelihood ratio test

Test statistic:

$$2\left(\log L_{full} - \log L_{reduced}\right)$$

Under $H_0$: $\sim \chi^2_q$ $q$ = # parameters tested

(or: null distribution from parametric bootstrapping)

# Likelihood ratio test

```
m2 <- zeroinfl(cigsPerDay ~ education +
                 diabetes + age | age,
               data = heart_data)
m2$loglik
```

```
## [1] -14023.42
```

```
m1 <- zeroinfl(cigsPerDay ~ education +
                 diabetes | age,
               data = heart_data)
m1$loglik
```

```
## [1] -14058.41
```

# Likelihood ratio test

```
m1 <- zeroinfl(cigsPerDay ~ education +
                 diabetes | age,
               data = heart_data)

m2 <- zeroinfl(cigsPerDay ~ education +
                 diabetes + age | age,
               data = heart_data)

2*(m2$loglik - m1$loglik)
```

```
## [1] 69.97593
```

```
pchisq(69.976, df=1, lower.tail=F)
```

```
## [1] 6.003042e-17
```

$\approx 0$

# Class activity

Work on the class activity. At the end of class, render your work as an HTML and submit on Canvas.